DASHing Factory: A Data-Driven Simulation and Emulation Platform for Optimizing 5G Industrial Applications

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***Abstract*—We introduces DASHing Factory, a simulation and emulation platform designed to evaluate the behavior of real-time applications, such as video streaming, in 5G-enabled factory automation. By integrating network simulation, link emulation, and data-driven approaches, the platform facilitates the generation of realistic network scenarios, the collection of performance data, and the application of machine learning (ML) and deep learning (DL) techniques for optimization. As a proof of concept, we focus on identifying network anomalies such as packet loss, delay, duplication, and reordering within a DASH-based use case for 5G factory automation. We tackle the problem from multiple angles, including binary classification, multi-class classification, and fine-grained detection of specific anomalies. The results demonstrate the platform's ability to provide valuable insights for adaptive streaming in industrial settings and highlight the potential of ML/DL techniques for real-time anomaly detection in 5G-enabled factory networks.**

***Index Terms*—5G Industrial Automation, Simulation and Emulation Platform, Anomaly Detection, AI-native Self-Organizing Networks, DASH**

I. Introduction

The increasing deployment of 5G networks in industrial settings is enabling a wide range of real-time applications, from video surveillance to robotic control. These mission-critical applications require high reliability, low latency, and the ability to adapt to fluctuating network conditions. This is a part of the larger transformation towards smart factories, where data-driven approaches are key to sustaining optimal performance and seamless operation. The application of machine learning (ML) and deep learning (DL) techniques for performance optimization is, therefore, receiving considerable attention. However, this interest underscores the need for platforms specifically designed to study and optimize these applications within 5G-enabled industrial environments.

To address this need, we present DASHing Factory, a comprehensive simulation and emulation platform. The platform integrates network simulation (via NS-3) and emulation (through Mininet) to replicate realistic 5G network conditions and generate detailed performance metrics under varying levels of congestion and impairments. A key feature of the platform is its ability to produce a rich dataset, capturing network conditions and performance metrics, providing the basis for data-driven techniques to optimize the overall system's behavior.

As a proof of concept, we implement an anomaly detection study focusing on network-level issues, such as packet loss, delay, duplication, and reordering. By utilizing ML and DL techniques, we demonstrate the platform's capability to detect and predict these anomalies in the context of a 5G-enabled Dynamic Adaptive Streaming over HTTP (DASH) use case. The significance of a platform extends into the evolving vision of *AI-native* networks, particularly within the paradigm of *Self-Organizing Networks (SON)*. In these networks, AI models autonomously configur, optimize, and even heal network functions in response to real-time conditions. Effective and timely anomaly detection is essential in this vision, enabling AI to proactively detect network disruptions, adjust signal strength, and optimize resource allocation. DASHing Factory supports this capability by offering a controlled environment to study and refine AI-driven strategies and thus facilitate the developpement of resilient and adaptable 5G networks that require minimal human intervention.

Finally, the platform's design is modular and versatile, extending beyond the studied use case. It can be applied to a wide range of protocols, network scenarios, and industrial applications, making it a robust tool for broader applications in 5G research.The contributions of this paper are twofold; First, the development of a robust simulation and emulation framework for 5G industrial application performance analysis; and second, the presentation of a comprehensive anomaly detection case study that demonstrates the platform's capabilities.

The remainder of this paper is organized as follows: We first reviews related work in Section II. In Section III, we provide a detailed description of DASHing Factory, outlining the platform's architecture and core components. Section IV, presents a comprehensive case study on anomaly detection in 5G factory automation, covering the use case, dataset, problem framing, methodologies, evaluation, and key findings. Finally, Section 5 concludes the paper with insights on future directions.

Table I
COMPARISON OF EXISTING TESTBEDS AND FRAMEWORKS (USE CASE, ML/DL INTEGRATION, AND RELEVANCE TO FACTORY AUTOMATION).

Ref.	Description	Use case studied	ML , DL	Factory Automation
[1]	Mahimahi : a framework to record traffic from HTTP-based applications, and later replay it under emulated network conditions.	Study of HTTP/1.1, SPDY, and QUIC under various em- ulated network conditions.	No	No
[2]	Speeder : framework for web performance measurement over many parameters (e.g., network conditions, browsers, metrics).	Evaluating how various best practices from HTTP/1 (concatenation, embedding, and sharding) apply to HTTP/2.	No	No
[3], [4]	goDASHbed : a framework using virtual environment for headless streaming of DASH video content locally through the Mininet virtual emulation tool. Used cellular trace datasets ([5]).	No use specefic use case studied.	No	No
[6], [7]	Data-driven framework with pre-installed DASH tools to analyse state-of-art Adaptive Bitrate Streaming (ABS) algorithms. Used 5G bandwidth values sampled from 5G traces.	Comprehensive comparison of HAS algorithms + ML classifier to predict user satisfaction based on network metrics, such as RTT, throughput, and number of packets.	DTR, MLR & RFR	No
[8]	QuicInteropRunner : a testing framework in which QUIC clients and servers interact with each other over a network that simulates various network conditions using ns-3.	Simple test cases where a client download a specific object from the server, receive a TLS session ticket or transert object.	No	No
[9]	Vegvisir : Automated testing framework that orchestrates prototypical QUIC-HTTP/3 client-server-based media streaming experiments over simulated network conditions.	Evaluation of HAS over HTTP/3	No	No

*DTR, MLR and RFR refer respectively to Decision Tree, Multi-linear and Random Forest Regression

II. RELATED WORKS

In surveying existing works, we find a range of testbeds and frameworks designed for network protocol testing, adaptive streaming, and latency analysis under varied network conditions, as summarized in Table I. While these platforms effectively address various network and application-layer scenarios, few integrate data-driven techniques to optimize system performance. Among these, studies in [6], [7] are the only to include the use of machine learning or deep learning techniques to demonstrate the potential for intelligent, real-time network management. Additionally, none of the surveyed works are specifically designed to meet the needs of factory automation or explicitly address the unique challenges of 5G-enabled industrial applications. This gap underscores the need for a versatile, data-driven testbed like DASHing Factory, which not only enables the simulation and emulation of complex network conditions but also supports the integration of ML/DL approaches for optimized, AI-native performance management in real-world industrial settings.

III. PLATEFORM DESCRIPTION

A. Architecture

Figure 1 provides a high-level architecture overview of the platform.

B. Components

1) Scenario Setup: The first step consists on defining the use case, network topology, parameters, and traffic patterns. Specifications from international standardization bodies,e.g., 3GPP, are a valuable resource for these settings.

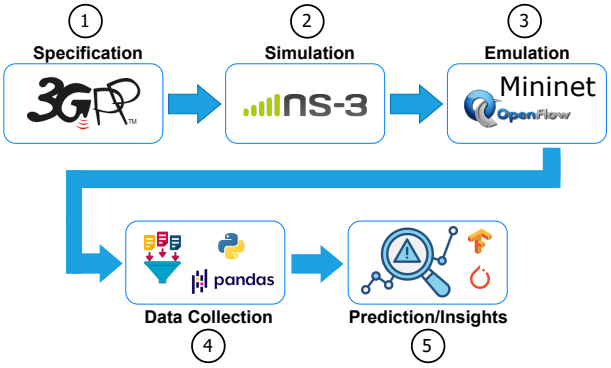

Figure 1. Outline of the DASHing Factory platform architecture : (1) Scenario Specification (3GPP-aligned), (2) Network Simulation (NS-3), (3) Network Emulation (Mininet, Linux TC, NetEm), (4) Data Collection and Processing (Pandas), and (5) Insights/Predictions (TensorFlow, PyTorch, Scikit-learn).

2) Network Simulation : The specified scenario is implemented using network simulators, e.g., NS-3 [10], to generate traces of key metrics such as throughput, delay, jitter, and packet error rate under varying levels of network conditions.

3) Link Emulation : Using the generated traffic traces, realistic network link conditions are emulated via tools like Mininet, alongside *Linux Traffic Control (TC)* and *NetEM*. These tools allow precise control over impairments such as packet loss, latency, and jitter, ensuring a realistic environment for testing real-world applications.

4) Data Collection : All network data, including application logs and packet capture files (pcaps), are collected and aggregated to create a comprehensive dataset. This dataset captures metrics such as throughput, delay, and loss rates, providing a foundation for subsequent analysis.

5) Insights/Predictions: The collected data is analyzed using ML/DL techniques to generate insights and make predictions, aiding in the optimization of the overall system under varying network conditions.

IV. Case Study: Anomaly Detection in 5G Factory Automation

A. Use Case Description

This study focuses on the 3GPP use case A.2.3.4: *Inspection in production systems* [11], where live video feeds from the production line enable remote supervision, allowing for quicker detection of issues and decision-making. Using NS-3, we simulate a factory layout aligned with 3GPP specifications, with a factory hall divided into 12 service areas, each containing up to 10 UEs. The network layout follows 3GPP guidelines, as shown in Figure 7.2-1 2 [12], and produces simulation outputs related to throughput under various levels of network congestion. The simulation setup uses 5G-LENA [13], a pluggable module to NS-3, used to simulate the New Radio (NR) aspects of the network [14], along with an NS-3 implementation of DASH [dash-ns3] to simulate the DASH client-server interaction of segment requests and response.

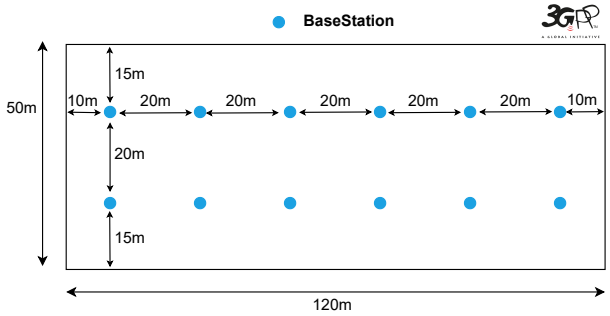

Figure 2. Platform layout and network configuration inspired by 3GPP specifications.

At the emulation stage, the NS-3 simulation traces are used to parameterize different link conditions in Mininet (via TC and NetEM). As in [6], Caddy, a WSGI-based web server, is used to host DASH video content, and goDash, a headless HTTP Adaptive Streaming (HAS) player, is used to replicate real-world streaming conditions. Both tools support a range of adaptive bitrate algorithms and metrics under varying link conditions (algorithm in Table III).

The approach enables the assessment of DASH performance under varying network conditions, including the injection of network anomalies such as packet duplication, reordering, and corruption, as outlined in Table II, applied with varying severity (5%, 10%, up to 50% of packets).

During data collection, logs and pcap files are aggregated into a single comprehensive dataset. In addition to network-level metrics, the dataset includes tags indicating the type and severity of the introduced anomalies.

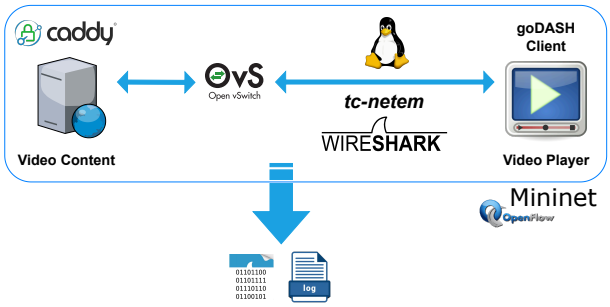

Figure 3. Overview of the emulation phase : Link condition adjustment using NS-3-derived 5G traces (applied via Linux TC/NetEM), real-world traffic generation (Caddy server and goDash player), and network impairment injection (e.g., delay, reordering).

Table II
Network impairments introduced during experiments including types and the respective impact on emulated traffic.

Impairement	Description
Packets Reordoring	Introduces out-of-order delivery of packets by holding some packets back for a specified amount of time. This can negatively impact protocols that rely on packet order (e.g., TCP), causing retransmissions and delays.
Packet Duplication	Packets are transmitted multiple times. This consumes extra bandwidth and may lead to confusion in the receiving end.
Packet Corruption	Introduces bit-level errors in the data, causing packets to be incorrect upon arrival. This could lead to retransmissions or, in worst cases, incorrect data being processed.

B. Dataset Overview

The resulting dataset contains more than 20 features, c.f., Table III, in addition to a target column that combines the following:

- *Anomaly class :* Four possible values — *reorder, duplicate and corrupt* (for respective anomalies), and *normal* (for normal cases).
- *Severity level :* Six possible values — 5%, 10%, 20%, 30%, 40%, or 50% of packets affected by the anomaly.

The dataset consists of 22,000 entries, with 21 columns, and is exported as a CSV file.

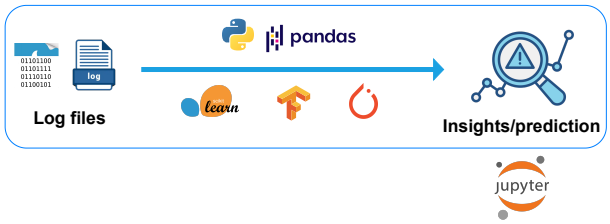

Figure 4. Pipeline for processing goDash logs, PCAP files, and per-segment QoS data to generate insights, including anomaly detection.

Table III
LIST OF DATASET METRICS WITH DESCRIPTIONS.

	Metric	Description
Per-Segment	Segment_no	Segment number *(60 segments streamed, of 2s each)*
	Seg Dur	Segment duration (2ms).
	Arr_time	Arrival time (ms).
	Del_Time	Delivery of the segment (ms).
	Rep_Level	Representation selected for the segment.
	Del_Rate	Delivery rate of network in kbps *(Byte Size $*$ 8 bits / Del Time)*.
	Act_Rate	Actual rate (kbps) *(Byte Size $*$ 8 bits / Seg Dur in seconds)*.
	Byte Size	Byte size of this segment.
Player-Features	Algorithm	Six variants of adaptive bitrate (ABR) algorithms tested : Rate-based *(Conventional and Exponential [15])*, Buffer-based *(Logistic [16]* and *BBA [17])* and Hybrid *(Arbiter [18]* and *Elastic [19])*.
	Width & Height	In pixels,
	Codec	Video encoder
	FPS	Frame rate of the streamed video
	Play_Pos	Current Playback position (ms)
	Stall_Dur	Stall or freeze duration (ms),
	Buff_Level	Buffer level in ms
Transp	RTT	Round-trip time at the packet level (ms) - determined using HTTP head request
	Protocol	HTTP protocol
QoE	P1203, Claey, Duanmu, Yin and Yu	QoE based on the model proposed in [20], [21], [22], [23] and [24], repectively.

C. Problem Statement

The goal is to develop models capable of discerning *normal* observations from various *anomalies*. This is addressed through multiple perspectives as depicted in Table IV.

Table IV
OVERVIEW OF THE CLASSIFICATION OBJECTIVES EXPLORED.

Approach	Description
Binary Classification	Determining the presence/absence of anomalies.
Multi-class Classification	Classifying anomalies into specific categories : *Reorder, Duplicate, Corrupt (including all levels of severity)* or *Normal* cases.
High-gravity Anomaly Detection	Detecting only *high-level (severity)* anomalies or the cases of *Normal* situations.
Fine-grained Anomaly Detection	Identifying the exact anomaly type along with the severity level.

D. Methodology

We approach the problem as a classification task using a variety of techniques. This includes Support Vector Machines (Kernel-based Methods), Multilayer Perceptron and TabNet (Feedforward and Attention-based Neural Networks), and XGBoost (Gradient Boosting methods), each offering distinct advantages/disadvantages (c.f.,V). The models then are evaluated based on two metrics: accuracy.

Training is performed on our dataset, with the goal of comparing the performance of these models across the different classification objectifs, c.f., (Table IV).

E. Evaluation

To ensure robust evaluation, we employed k-fold cross-validation, where the dataset is divided into 'k' subsets of equal size. The value for 'k' is fixed to 10, a commonly used choice in practice [31]. The outcomes of our experiments are depicted in Table VI, illustrating the performance of ML models across different classification problems. Figure 5 illustrates the results in termes of accuracy.

F. Discussion and Basic Findings

The analysis of the previous results yields several insights :

1) Performance in Classification Tasks : The models performed very well in both binary, multi-class and high gravity classification tasks. Specifically:

a) Binary Classification : achieved an average accuracy of 93%, demonstrating strong ability to distinguish between normal and abnormal behavior.

b) Multi-class Classification : the models achieved an average accuracy of 85%, successfully categorizing anomalies into their respective classes *(reorder, duplicate, corrupt)*.

c) High-gravity Anomaly Detection : the models excelled in identifying severe anomalies, with an average accuracy of 95%.

d) Fine-Grained Anomaly Detection: performance significantly dropped in the fine-grained detection task, with an average accuracy of 59% (including a relatively good performance of 88% and 76% for XGBoost and TabNet respectively). This suggests the inherent complexity of this task, as it requires identifying not only the anomaly type but also its severity level. One key challenge is the detection of low-severity impairments, such as anomalies affecting only 5% of packets, often misclassified as normal, resulting in higher false positive rates.

2) Anomaly Type Detection: To further analyze the models' incapacity of detecting specific impairments, we examined the F1-scores for each anomaly type in the context of fine-grained anomaly detection, c.f. Table VII :

a) Corrupt Anomalies : Across all models, corrupt anomalies were consistently easier to detect compared to reorder and duplicate cases. The nature of corrupt packets might introduce more noticeable patterns that are easier for the models to learn.

b) Reorder vs. Duplicate : Between reorder and duplicate anomalies, there were variations in performance across models: For neural network-based approaches, reorder anomalies were relatively easier to detect. For gradient boosting and kernel-based approaches (e.g., SVM), reorder anomalies proved to be the most challenging. This suggests that certain model architectures are more sensitive to specific patterns in network impairments, with reorder anomalies posing particular difficulties for some models.

Table V
SUMMARY OF ALGORITHMS USED IN THE EVALUATION, THEIR ADVANTAGES AND LIMITATIONS.

Technique	Familly	Description	Advantages	Disadvantages
SVM	Kernel-based Methods	A supervised ML algorithm particularly effective for classification tasks. The main objective is to find a hyperplane that best separates data points belonging to different classes in a high-dimensional space [25], [26].	Can handle cases where the number of dimensions exceeds the number of samples, Effectiveness in cases where data is not linearly separable, Generalization to unseen data.	Scalability, Requires careful tuning of parameters (regularization, kernel). Interpretability.
MLP	Feedforward Neural Networks	Neural Network is a computational model inspired by the structure and functioning of the human brain. They are trained using data, adjusting the weights based on the difference between predicted and actual outcomes. MLP is a simple type of neural network where each layer is fully connected to the next one [27].	Usability (used for a wide range of tasks), Scalability, Ease of implementation.	Interpretability, Prone to overfitting, Computationally intensive.
TabNet	Attention-based Neural Networks	A more complex Neural Network architecture using sequential attention, particulary effective for tasks involving structured (tabular) data, combining interpretability of decision trees with the power of deep learning [28], [29].	Interpretability, designed to handle tabular data efficiently, feature selection.	More complex to implement and tune (compared to decision trees or MLPs), Can be computationally demanding.
XGBoost	Gradient Boosting Machines	An advanced and highly efficient implementation of gradient boosting, consists on buildings an ensemble of trees sequentially, where each new tree corrects the errors of the previous ones, designed for speed and performance making it a popular choice for structured (tabular) data tasks [30].	Performance and speed (often achieving state-of-the-art results), Feature importance.	Can be complex to tune and requires careful parameter optimization, Prone to overfitting, Resource intensive.

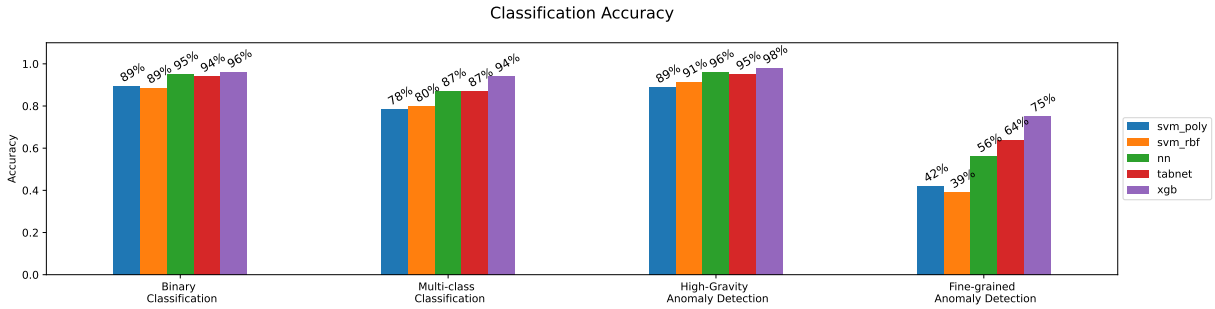

Figure 5. Comparative performance of classification algorithms for anomaly detection, measured by accuracy.

Table VI
ACCURACY RESULTS ACROSS BINARY, MULTI-CLASS, HIGH-GRAVITY, AND FINE-GRAINED ANOMALY DETECTION, INCLUDING AVERAGE ACCURACY PER CLASSIFICATION TASK.

	SVM		Neural Networks		XGboost	AVG (%)
	Poly	RBF	MLP	TabNet		
Binary classification	0.895	0.886	0.949	0.94	0.96	93%
Multi-Class Classification	0.784	0.799	0.87	0.87	0.94	85%
High-Gravity Anomaly Detection	0.902	0.921	0.979	0.971	0.985	95%
Fine-Grained Anomaly Detection	0.492	0.473	0.59	0.647	0.759	59%

Table VII
F1-SCORES FOR EACH SPECIFIC ANOMALY TYPE IN THE CONTEXT OF FINE-GRAINED ANOMALY DETECTION, ILLUSTRATING MODEL PERFORMANCE PER ANOMALY CATEGORY.

	SVM		Neural Networks		XGboost	AVG (%)
	Poly	RBF	MLP	TabNet		
Normal	0.69	0.65	0.85	0.88	0.93	80%
Reorder anomalies	0.12	0.065	0.392	0.52	0.573	33%*
Duplicate Anomalies	0.277	0.276	0.32	0.416	0.61	38%**
Corrupt Anomalies	0.65	0.591	0.672	0.76	0.88	71%

* 9% for SVM (poly, RBF) and 45.6% for NN (MLP, TabNet) based mothods
** 27.6% for SVM (poly, RBF) and 36.8% for NN (MLP, TabNet) based methods

These findings highlight both the potential and limitations of current anomaly detection techniques in this domain. The higher detection accuracy for corrupt anomalies suggests that certain types of impairments are more distinguishable than others. Future work will focus on improving fine-grained detection by refining model architectures and exploring feature engineering techniques tailored to subtle network impairments.

V. CONCLUSIONS

In this work, we present DASHing Factory, a platform that combines network simulation, emulation, and data collection

to study the performance of DASH in 5G factory automation scenarios. By leveraging NS-3 for simulation and Mininet for emulation, we were able to create a realistic and controlled environment for studying the impact of network conditions on video streaming quality. As a proof of concept, we performed an anomaly detection analysis using ML and DL models to detect network anomalies like packet loss, delay, duplication, and reordering. Our results show that while binary classification and multi-class classification tasks performed well, fine-grained anomaly detection remains a challenge, especially for low-severity anomalies.

While anomaly detection for video streaming serves as a proof of concept in this study, the DASHing Factory platform is versatile and adaptable, offering a reusable framework for evaluating various protocols and use cases in industrial environments. Future work will focus on expanding the platform to incorporate real-world factory setups and other adaptive streaming technologies which could further enhance its utility for industrial automation and 5G research.

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